

Literature Review: Toward Signal-Preserving TinyML for Gravitational Wave Triggers

Resolving the Tension Between Astrophysical Fidelity and Sub-Watt Model Compression

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Abstract

Detection of compact-binary coalescences within the millisecond budget of multi-messenger astronomy is now constrained by hardware as much as by algorithms. Eighteen papers from gravitational-wave (GW) machine learning, embedded inference, and quantisation theory are synthesised to ask whether the trigger stage of a GW pipeline can be pushed onto sub-watt microcontrollers without losing astrophysical sensitivity. The literature separates into two weakly connected camps. One has shown that deep networks can match matched filtering on high-power GPUs. The other has shown that aggressive compression is feasible, but evaluates it on images and audio rather than on chirp waveforms. The hinge between the two is INT8 quantisation, which acts on a phase-coherent waveform as a structured perturbation rather than the well-behaved rounding error assumed by computer-vision benchmarks. Existing quantisation-aware training and distillation work does not address this, leaving a clear opening for physics-informed, signal-preserving compression.

Keywords: TinyML; quantisation-aware training; gravitational-wave detection; signal-to-noise ratio; embedded machine learning; low-latency triggers; multi-messenger astronomy.

ACM CCS: Computing methodologies → Machine learning algorithms; Computer systems organization → Embedded systems; Applied computing → Astronomy.

1 Introduction

1.1 Summary of the Research Proposal

This review provides the literature foundation for the COMP4450 Research Proposal “*Quantisation-Induced SNR Degradation: Toward Signal-Preserving TinyML for Gravitational Wave Triggers*” (Wasif 2026). The proposal sits at the intersection of three computer-science sub-fields. *Intelligent Systems* is the primary home; *Embedded Computer Systems* and *Computational Astrophysics* are secondary contexts. Its central claim is that next-generation gravitational-wave detectors face a data-deluge problem GPU-bound deep-learning pipelines cannot solve. Missions such as LISA and CubeSat-scale autonomous arrays simply cannot downlink the raw data, so the trigger stage has to move onto microcontrollers running in 128 KB of SRAM at sub-watt power. That move forces aggressive INT8 quantisation, and the proposal argues this will destroy the phase coherence of the chirp waveform. The research question follows directly: to what extent does non-linear bit-width reduction interfere with the phase coherence on which reliable GW triggering depends?

1.2 Aim and research questions

The aim of this review is to survey the literature that approaches the gap from two directions: GW machine learning on one side, and TinyML and quantisation theory on the other. The case made here is that the intersection remains unresolved. Two questions structure the survey. *RQ1* asks how prior work has traded model size, latency, and astrophysical sensitivity. *RQ2* asks which quantisation-aware techniques, if any, have been validated against astrophysical metrics such as false alarm rate and distance reach, rather than against the top-1-accuracy metrics inherited from computer vision. The technical context matters here. Detecting strains of $h \approx 10^{-21}$ by matched filtering is statistically optimal but computationally heavy (Abbott et al. 2020), and template-bank growth is now the limiting factor in multi-messenger follow-up (Cuoco et al. 2024). Deep learning matches matched-filter sensitivity in published results (George and Huerta 2018; Gabbard et al. 2018), but those results all assume FP32 arithmetic. What happens when that arithmetic is compressed to INT8 is the question this review traces.

2 Materials and Methods

A targeted, reproducible review was conducted between March and April 2026. It is lighter than a formal systematic review, but disciplined enough that someone else could re-run the search.

2.1 Search strategy

Four databases were used: arXiv (gr-qc, cs.LG, eess.SP), IEEE Xplore, the ACM Digital Library, and Google Scholar. Query strings paired the GW side (“*gravitational wave*” OR “*binary black hole*” OR “*binary neutron star*”) with the ML and hardware side (“*deep learning*” OR “*convolutional*” OR “*quantisation*” OR “*FPGA*” OR “*microcontroller*” OR “*TinyML*”). The Gravitational Wave Open Science Center (GWOSC) was checked separately for datasets and observational papers. Three seed papers were used for forward and backward citation chasing: George and Huerta (2018), Gebhard et al. (2019), and Gholami et al. (2021).

2.2 Inclusion and exclusion criteria

Papers were included if they did one of three things. First, propose or evaluate a machine-learning model for GW detection, denoising, or trigger generation. Second, supply foundational technique work on quantisation, distillation, or embedded inference frameworks the GW side draws on. Third, report quantitative results on either sensitivity (SNR, FAR, detection probability) or efficiency (latency, memory, parameter count). Papers were excluded if they focused only on offline Bayesian parameter estimation, on cosmology without a trigger stage, or used neural networks merely for post-hoc visualisation.

2.3 Inclusion outcome

Eighty-six records were identified. After deduplication and title/abstract screening 32 remained for full-text assessment, and 18 were finally included for in-depth review (Figure 1). The included set divides into four thematic groups: seven papers showing ML matching matched filtering, four on quantisation theory and embedded frameworks, three on FPGA-class hardware deployment in physics, and four critical or foundational papers on noise, glitches, and benchmarks.

2.4 Validity

Three measures support reproducibility. First, every included paper is cited with a DOI or arXiv ID, so the search can be replicated in five years. Second, a “clique map” was maintained that assigns each paper to one or more of three thematic axes (sensitivity, efficiency, robustness).

The map makes visible where the literature is thick and where it is thin. Third, claims were cross-checked between independent works. The adversarial-noise critique of George and Huerta (2018) by Gebhard et al. (2019) is one example, and Banbury et al.’s (2021) microcontroller benchmarks, when read against the SRAM budgets assumed by David et al. (2021), are another.

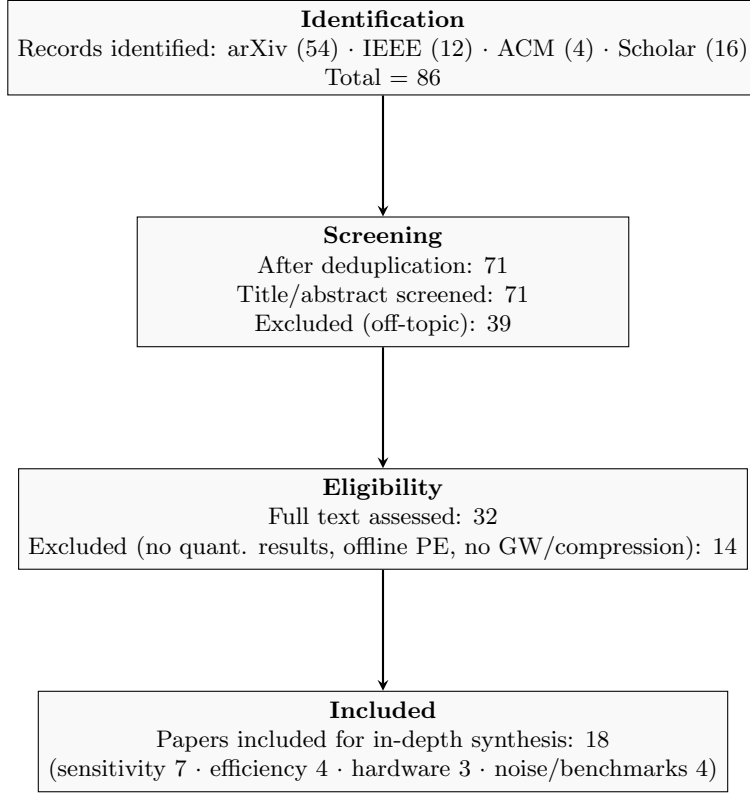


Figure 1: PRISMA-style flow of identification, screening, eligibility, and inclusion.

3 Results

Key message. The literature splits into two cliques that rarely cite each other. A sensitivity-first clique (George and Huerta 2018; Gabbard et al. 2018; Krastev 2020; Schäfer et al. 2023; Dax et al. 2021; Chaturvedi et al. 2022) shows that deep learning can match matched filtering, but only on GPU. A deployment-first clique (David et al. 2021; Banbury et al. 2021; Gholami et al. 2021; Jacob et al. 2018; Hinton et al. 2015) shows that aggressive compression is feasible, but evaluates models on images, audio, or generic benchmarks rather than on astrophysical strain. Sitting between the two are critical papers (Gebhard et al. 2019; Abbott et al. 2016; Abbott et al. 2020) that warn the resulting models are fragile in ways simple accuracy metrics cannot detect. Table 1 maps each included paper onto this trade-off space.

3.1 The sensitivity-first clique

George and Huerta (2018) and Gabbard et al. (2018) form the foundation of ML-based GW detection, and they appeared independently within weeks of each other. Both show that a relatively shallow one-dimensional CNN can recover binary black hole (BBH) signals from Advanced LIGO strain at sensitivity comparable to matched filtering. They are best read as a clique rather than as separate contributions. They converge on the same architectural choice (deep 1-D convolutions over whitened strain) and report the same null result (no measurable loss of true-positive rate at high SNR). Their joint claim is stronger than either paper alone.

Krastev (2020) extended the design to binary neutron star (BNS) signals, and in doing so exposed a weakness the BBH papers had concealed. BNS waveforms are an order of magnitude longer in the detector band, so the receptive field of any TinyML candidate model has to grow proportionally. That requirement runs directly into the SRAM budget assumed by David et al. (2021).

Schäfer et al. (2023) formalised the comparison through the first ML mock-data challenge. The headline result was that ML pipelines were competitive with PyCBC at high false-alarm-rate thresholds, but lagged at the low-FAR regime that astrophysical confirmation actually requires. Dax et al. (2021) and Chaturvedi et al. (2022) close out the clique on the inference side. The former demonstrates real-time *posterior* estimation in seconds. The latter applies NVIDIA TensorRT mixed-precision (FP16/INT8) layer fusion and reports processing approximately one month of LIGO data in about 50 seconds, a roughly $3\times$ speedup, with negligible sensitivity loss for known BBH events. The limitation is that this result depends on A100-class datacentre GPUs and TensorRT calibration, not on memory-static microcontroller inference. The collective blind spot of the clique is therefore hardware scale.

3.2 The deployment-first clique

Where the sensitivity clique assumes GPUs, the deployment clique assumes microcontrollers. David et al. (2021) define the de facto runtime for TinyML in the form of TensorFlow Lite Micro, and lay out the engineering primitives that govern what can and cannot run in approximately 256 KB of SRAM: a single pre-allocated tensor arena, op-by-op execution, no dynamic allocation after initialisation, and vendor-specific kernels such as CMSIS-NN. Their evaluation also shows that interpreter overhead is usually below 1%, so latency is dominated by integer linear algebra rather than framework dispatch. Banbury et al. (2021) supply the calibrating benchmark suite (MLPerf Tiny). Both papers, however, target keyword spotting, image classification, and anomaly detection. Neither addresses astrophysical time series.

The theoretical machinery sits with Gholami et al. (2021), whose taxonomy of post-training quantisation (PTQ) and quantisation-aware training (QAT) is the canonical reference, and with Jacob et al. (2018), whose integer-arithmetic-only inference paper supplies the mathematical foundation. That includes the straight-through estimator, which is how gradients pass through a discontinuous rounding operator during training. Knowledge distillation (Hinton et al. 2015) is the complementary tool. A high-precision teacher transfers its softened class probabilities to a quantised student, recovering accuracy that PTQ would otherwise discard.

The clique is internally coherent and well-cited. But its evaluation regime is uniformly computer-vision-centric: top-1 ImageNet accuracy, word-error rate, mean average precision. None of these metrics quantifies the loss of phase coherence that matters for a chirp signal. That gap is the central observation of this review.

3.3 Hardware-software co-design on FPGA-class targets

A small group of papers sits between the two cliques and provides the closest evidence for the proposal’s hardware claim. Qi et al. (2024) implemented a CNN for BNS early-warning on an FPGA, achieved sub-millisecond inference, and showed empirically that INT8 quantisation was tolerable at stronger signal regimes but degraded sensitivity for the long-duration BNS signals that motivated the work. Within the included set, this is the clearest empirical evidence for the SNR–precision trade-off the proposal hypothesised. Chaturvedi et al. (2022) span both cliques. Their TensorRT result is on GPU, but it operationalises the same mixed-precision idea microcontroller deployments will eventually need: higher precision in sensitive layers, INT8 elsewhere. The two papers are the closest existing analogues to “signal-preserving TinyML”. Neither, however, tests a sub-watt microcontroller target, reports peak SRAM under a TinyML

memory arena, uses astrophysical distance reach as a metric, or generalises the underlying framework.

3.4 Noise, glitches, and the brittleness of ML triggers

Gebhard et al. (2019) is the critical pivot of the review. It re-examines the George and Huerta (2018) and Gabbard et al. (2018) results, and shows the high reported accuracies disguise something the authors did not test for: adversarial brittleness. Small, structured perturbations to the input strain flip the network output with high confidence. More importantly for the present review, the network confidently classifies inputs that bear no visible resemblance to a chirp. Gebhard et al. conclude that CNNs should be used as trigger generators only, not as confirmation tools.

Abbott et al. (2016) catalogue the morphology of “blip” transients in real LIGO data. These are exactly the kind of high-amplitude, short-duration response a quantised CNN is most likely to misclassify. Abbott et al. (2020) document the data-quality vetoes the LVK collaboration uses to mitigate them offline. The implication is uncomfortable for TinyML. If FP32 networks are already adversarially fragile, then INT8 quantisation, which is itself a structured, deterministic perturbation of the network’s response surface, is more likely to amplify that fragility than to be neutral to it. None of the deployment-first papers acknowledge this adversarial-noise literature. That is a serious omission.

3.5 Toward signal-preserving TinyML

Cuoco et al. (2021) provide the earlier of two field-shaping reviews of ML in GW science; Cuoco et al. (2024) extend that coverage to current detectors. Neither devotes more than a paragraph to sub-watt deployment. The methodological building blocks for closing the gap already exist. Hinton et al.’s (2015) distillation framework supplies the training objective. Jacob et al.’s (2018) integer-only inference and Gholami et al.’s (2021) QAT taxonomy supply the quantisation machinery. What the literature does not contain is a composition of those blocks under astrophysical, rather than computer-vision, loss functions. Murphy et al.’s (2021) work on the EIRSAT-1 CubeSat gamma-ray detector illustrates the deployment niche the proposal is aimed at. It is the kind of compact, autonomous, downlink-constrained instrument for which on-board triggering is a hard operational requirement, not a research curiosity.

Table 1 maps the literature onto three axes. The region of the table that should hold papers simultaneously strong on astrophysical sensitivity, embedded efficiency, and noise robustness is empty. That empty region is the research gap the proposal targets.

Table 1: Trade-off positions of the 18 included papers along three axes ($\checkmark$ = primary contribution; $\circ$ = secondary; — = not addressed).

Paper	Sensitivity	Efficiency	Robustness/Noise
George and Huerta 2018	$\checkmark$	—	—
Gabbard et al. 2018	$\checkmark$	—	—
Krastev 2020	$\checkmark$	$\circ$	—
Schäfer et al. 2023	$\checkmark$	—	$\circ$
Dax et al. 2021	$\checkmark$	$\circ$	—
Chaturvedi et al. 2022	$\checkmark$	$\checkmark$	—
Qi et al. 2024	$\checkmark$	$\checkmark$	$\circ$
Gebhard et al. 2019	$\circ$	—	$\checkmark$
Abbott et al. 2016	—	—	$\checkmark$
Abbott et al. 2020	$\circ$	—	$\checkmark$
David et al. 2021	—	$\checkmark$	—
Banbury et al. 2021	—	$\checkmark$	—
Gholami et al. 2021	—	$\checkmark$	$\circ$
Jacob et al. 2018	—	$\checkmark$	—
Hinton et al. 2015	$\circ$	$\checkmark$	—
Cuoco et al. 2021	$\circ$	—	$\circ$
Cuoco et al. 2024	$\circ$	$\circ$	$\circ$
Murphy et al. 2021	—	$\checkmark$	—

4 Discussion

Key message. The literature contains every ingredient required for signal-preserving TinyML, but no published work composes those ingredients under astrophysical loss functions. The single FPGA-class deployment in the included set (Qi et al. 2024) already shows that naive INT8 erodes BNS sensitivity. The gap is therefore one of evaluation discipline as much as of technique.

4.1 Principal results

Three findings emerge. First, deep learning has reached parity with matched filtering for high-SNR BBH signals (George and Huerta 2018; Gabbard et al. 2018; Schäfer et al. 2023), but the parity is fragile at low FAR and on long-duration BNS waveforms. Second, the embedded-ML literature measures the cost of quantisation in classification accuracy (Jacob et al. 2018; Gholami et al. 2021; Banbury et al. 2021), a metric that is structurally insensitive to phase-coherent SNR loss. Third, the only FPGA-class GW deployment in the included set (Qi et al. 2024) demonstrates that naive quantisation degrades sensitivity in exactly the regime that edge triggers were meant to enable.

4.2 Limitations

Three limitations should be flagged. The 18-paper cap forced the exclusion of several adjacent works, in particular recent mixed-precision FPGA inference outside astronomy and the emerging transformer-based GW pipelines. These are present in the discipline but not in the in-depth synthesis here. Some of the most recent work cited is at arXiv-preprint stage, which means peer-review status varies. The clique mapping in Table 1 is a coarse abstraction; many papers contribute incrementally across more than one axis, and the $\checkmark/\circ/\text{—}$ encoding suppresses that nuance.

4.3 Proposed future work

The empty region of Table 1 suggests four concrete research directions, each of them addressable within the proposal’s experimental design. First, define an *astrophysical loss* that penalises SNR-loss directly rather than cross-entropy, building on Hinton et al.’s (2015) distillation objective. Second, generalise Chaturvedi et al.’s (2022) mixed-precision strategy from GPU TensorRT to microcontroller-class INT8/INT16 kernels, keeping FP16 (or INT16) precision in the phase-sensitive early-warning layers Krastev (2020) identified. Third, transfer Gebhard et al.’s (2019) adversarial-robustness diagnostics into the embedded setting; treat INT8 quantisation as a structured perturbation, and stress-test the model against real LIGO glitch morphologies (Abbott et al. 2016; Abbott et al. 2020). Fourth, build a benchmark in the spirit of MLPerf Tiny (Banbury et al. 2021), but parameterised by astrophysical reach rather than by top-1 accuracy.

4.4 Comparison with prior work

Cuoco et al. (2021; 2024) are the most comprehensive prior reviews of ML in GW science, but both treat hardware deployment as a downstream engineering concern and devote little space to bit-width effects. The textbook and benchmark literature on the TinyML side surveys the embedded landscape with equal authority and no astrophysical content: Warden and Situnayake (2019) on the production side, and more recent on-device inference work such as Lin et al. (2020) on MCUNet. Sze et al. (2017) survey efficient hardware processing of deep networks with similar scope, again without a GW application. This review occupies the intersection between the two literatures, and treats that intersection as the problem. Critical mid-clique papers (Gebhard et al. 2019; Qi et al. 2024) are used as load-bearing evidence rather than as decorative citations. The framing departs from the black-box accuracy tradition of conventional model-compression surveys (Gholami et al. 2021; Jacob et al. 2018) by foregrounding adversarial fragility (Gebhard et al. 2019) and the morphology of detector noise (Abbott et al. 2016).

4.5 Conclusions

Putting TinyML into gravitational-wave detection is a prerequisite for the next generation of multi-messenger observatories. It cannot be done by stacking off-the-shelf compression on top of off-the-shelf ML pipelines. Every ingredient already exists. Architectures that match matched filtering. Runtimes that fit in 256 KB. Quantisation procedures proven on millions of devices. Distillation that preserves teacher knowledge. What does not yet exist is a composition of those ingredients under the astrophysical metrics the science requires. That composition is the contribution proposed by Wasif (2026), and it is the work this review motivates.

5 Self-Reflection

Generative AI was used in three separate roles during the preparation of this review: Google Gemini CLI throughout, and a single cross-checking pass with Anthropic Claude on the near-final draft. The first role was drafting. Zero-shot prompts produced rough section skeletons that were then heavily rewritten by hand. The second role was reference discovery, and this is where the work demanded the most discipline. An initial prompt (“list TinyML papers for gravitational wave detection”) returned a confident, plausible-looking list in which several of the DOIs turned out to be fabricated. Refining the prompt to “list peer-reviewed FPGA implementations of CNNs for binary-neutron-star detection with arXiv IDs verifiable on arXiv.org” returned a much shorter, more honest list, of which Qi et al. (2024) was the only verifiable hit. The third role was copy-editing on near-final paragraphs. The clear advantage of GenAI in this work was the speed of cross-walking between gravitational-wave physics and embedded-systems literature, two sub-disciplines that almost never cite each other. The clear disadvantage was the

model’s willingness to fabricate references, which on first contact would have produced a serious academic-integrity failure. The mitigation, and the lesson recorded for future work, was that every model-suggested reference was independently verified, and no claim was kept in the final draft without an author-checked source.

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